



Liquid-liquid equilibrium for n-hexane + benzene + sulfolane, + 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide ([EMIM][NTf₂]), + 1-ethyl-3-methylimidazolium ethylsulfate ([EMIM][EtSO₄]) and + the mixtures of [EMIM][NTf₂] and [EMIM][EtSO₄]

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ABSTRACT

The liquid-liquid equilibrium (LLE) data for three ternary mixtures of n-hexane + benzene + sulfolane, + 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide ([EMIM][NTf₂]) and + 1-ethyl-3-methylimidazolium ethylsulfate ([EMIM][EtSO₄]) and two quaternary ones of n-hexane + benzene + mixed ionic liquids respectively with fixed molar ratio 9:1 and 1:9 of [EMIM][NTf₂] to [EMIM][EtSO₄] were determined experimentally at (303.15, 313.15 and 323.15) K and atmospheric pressure. The experimental results showed that the ionic liquid of [EMIM][NTf₂] captures higher molar distribution coefficient while one of [EMIM][EtSO₄] is characterized with higher molar selectivity, compared to the sulfolane solvent that is extensively used for aromatic extraction in industry. The improvement of the composite separation performance for n-hexane and benzene was observed through mixing the above two ionic liquids. To represent the experimental diagram, the activity coefficient model of Non-Random Two-Liquid (NRTL) was employed to correlate the experimental data and good agreements between theoretical and experimental results were obtained. In general, the selected ionic liquid and its mixture in this work could be considered as a potential substitute for sulfolane.

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1. Introduction

The aromatic substances like benzene, toluene and so on are very important raw materials. They are widely used for the production of various petrochemical products. However, the separation of aromatic hydrocarbons and alkanes is accompanied with huge energy consumption and serious environmental pollution [1,2]. Furthermore, their extremely closing boiling point and/or forming azeotropes make the process more difficulty. To overcome the defects, the liquid-liquid equilibrium extraction technology has been extensively applied in the industrial practice nowadays, since the process captures an easy operation, high efficiency, low energy consumption and cost [3]. For aromatic purification, the most com-

monly used extractant in industrial processes is sulfolane [4–7] at present. While the ionic liquids (ILs), those were considered as a kind of ideal replacement for traditional volatile solvents due to their non-volatile, thermally stable, non-flammable, recyclable, and reusable characteristics, have also been getting more and more attentions.

So far, the ionic liquids have been extensively employed for separating different systems, especially for azeotropic mixtures as ones summarized by Pereiro et al. [8]. Among these researches, much work has been contributed to the separation of aromatics and alkanes. For instance, many efforts have been carried out for aromatic separation based on ILs by Arce et al. [9–12], Domańska et al. [13–15], Garcia et al. [16–18], González et al. [19–23], Larriba et al. [24–28], Meindersma et al. [29–32] and so on. In the above literatures' work, the investigated ILs solvents were mainly consisted of the imidazolium-based cations and the anions like bis(trifluoromethylsulfonyl)imide ([NTf₂][−]), methylsul-

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fate ([MSO₄][−]), ethylsulfate ([EtSO₄][−]), thiocyanate ([SCN][−]), etc. There are of course other ILs being used for liquid-liquid extraction separation of aromatic products. For instance, the ionic liquids formed by the cations of 1-ethyl-3-methylimidazolium ([EMIM]⁺), 1-butyl-3-methylimidazolium ([BMIM]⁺), 1-octyl-3-methylimidazolium ([OMIM]⁺) and 1-butyl-4-methylpyridinium ([4bmpy]⁺) and the anions of tricyanomethanide ([TCM][−]), dicyanamide ([DCA][−]), tetrafluoroborate ([BF₄][−]) and hexafluorophosphate ([PF₆][−]) [25,33,34] were selected as extractive solvents for the separation of benzene or toluene from alkanes. The results showed that the distribution coefficient and selectivity for toluene by [4bmpy][TCM] and [EMIM][TCM] are higher than those of sulfolane, and the separation factor of [EMIM][DCA] was even twice one of sulfolane. [EMIM][TCM] was considered as a more suitable substitution for sulfolane due to its appropriate density and viscosity [27]. Furthermore, the explorations about the mixing ionic liquid solvents for aromatic separation have also been performed. In García et al.'s work [35], the binary mixture of [4bmpy][NTf₂] and [EMIM][EtSO₄] was made to be as a solvent for separating toluene from heptane through liquid-liquid extraction. The experimental results showed that the extraction ability for toluene is higher than that of sulfolane when the mixing ionic liquid solvent has an equimolar composition. The mixture solvent of [4bmpy][NTf₂] and [EMIM][DCA] [36] were also used to separate toluene from alkanes and the experimental liquid-liquid equilibrium (LLE) results at 313.15 K and atmospheric pressure demonstrated that both the distribution and selectivity coefficients are higher than those of sulfolane when the molar fraction of [4bmpy][NTf₂] is 0.3 in the mixed solvent.

Among ionic liquid solvents for aromatic separation, the anions like [NTf₂][−] and [EtSO₄][−] have been paid much more attentions due to their capturing similar functional group with sulfolane. They have also been verified to be positive in improving the separation of aromatics. The liquid-liquid equilibrium data of hexane + benzene + [EMIM][NTf₂] at 298.15 K and 313.15 K suggested that benzene could be effectively extracted from hexane, and [EMIM][NTf₂] could be a substitute for traditional extractant compared with sulfolane and other ionic liquids [11]. What's more, the investigation about the effect of cations in [C_nMIM][NTf₂] on separation performance of benzene and hexane through LLE measurements at 298.15 K exhibited that the short alkyl chain contained in cation captures better separation ability [10]. In addition, the LLE data at 298.2, 303.2 and 328.2 K showed that [EMIM][EtSO₄] has a good selectivity for benzene [16], which has also been verified by González et. al. [20] with LLE data at 298.15 K. From the above investigation of LLE data for n-hexane + benzene + ILs, one can observe that [EMIM][NTf₂] and [EMIM][EtSO₄] respectively possess better solubility or selectivity for benzene compared to sulfolane solvent [37], while the sulfolane is still the best solvent in balancing both soluble and selecting capacities. So, the study on ILs in separating aromatics is still incomplete. In this work, the separation and dearomatics abilities of [EMIM][EtSO₄], [EMIM][NTf₂] and their mixtures with different mixing molar ratios will be further tested. The sulfolane was selected as a baseline to validate the separation performance of ILs. Benzene and n-hexane were used to represent aromatics and alkanes respectively. The LLE measurements for different ternary and quaternary mixtures at 303.15, 313.15 and 323.15 K and atmospheric pressure were firstly carried out. The Non-Random Two-Liquid (NRTL) was then employed to correlate the LLE data. The detailed discussions about the solubility and selectivity of sulfolane and ILs for benzene were made at last and the feasibility of mixed ILs as the separation extractant of aromatics and alkane was verified.

2. Experimental sections

2.1. Chemicals

The chemicals of n-hexane, benzene, ethanol and sulfolane were purchased from Sinopharm Chemical Reagent Co., Ltd. The ionic liquids of [EMIM][NTf₂] and [EMIM][EtSO₄] were from Shanghai Moni Chemical Technology Co., Ltd. The purity of the chemicals except ILs was checked by a gas chromatograph (GC). All chemicals were used without further purification except that the IL samples were evaporated in a rotating evaporator for at least 4 hours and dried in a vacuum dryer (pressure 0–5 kPa) for at least 24 hours before used. The detailed information about the chemicals were listed in Table 1, in which the chemical name, CAS No., the purity, water content and so on were included.

2.2. Apparatus and procedure

The LLE data of different mixtures were measured at 303.15, 313.15, and 323.15 K and atmospheric pressure respectively. The selection of temperature and pressure conditions in this work was based on the range of conditions commonly used for the separation of alkanes and aromatics by ILs [29,30]. The experiment was carried out in a self-made glass phase equilibrium cell with a constant temperature jacket, which was connected with a constant temperature (±0.1 K) water bath. First of all, the samples were accurately prepared with the electronic analytical balance (±0.1 mg) and then add them to the kettle. After sealing, the mixture was stirred for 2 h under the set temperature and then left to settle at least 3 h to ensure thermodynamic equilibrium and complete phase separation. At last, both samples were taken from two equilibrium liquid phases respectively. To get the equilibrium composition, the ethanol was used as an internal standard and the content of ILs was calculated through the method of mass conservation. Each sample was quantitatively analyzed by gas chromatography (PANAN A91) at least 3 times until the deviation of liquid composition was less than 0.0010 mole fraction. In gas chromatograph analysis, the FID detection and PONA chromatographic column (50 m × 0.2 mm) were used with the carrier gas of nitrogen.

3. Results and discussion

3.1. Experimental data

The LLE data for the mixtures of n-hexane (1) + benzene (2) + sulfolane (3), + [EMIM][EtSO₄] (3), + [EMIM][NTf₂] (3), + mixed ILs (3) with molar ratio 1:9 and 9:1 of [EMIM][EtSO₄] to [EMIM][NTf₂] at 303.15, 313.15 and 323.15 K and atmospheric pressure were respectively listed in Tables 2–6, in which the temperatures, the equilibrium compositions of raffinate and extraction phases and distribution and selection coefficients were included. The distribution and selection coefficients are key parameters of liquid-liquid extraction process, and the former is directly related to the solvent ratios while the latter determines the separation stages. They were calculated by

$$K_i = x_i^E / x_i^R \quad (1)$$

and

$$S_{ij} = \frac{K_i}{K_j} = \frac{x_i^E / x_i^R}{x_j^E / x_j^R} \quad (2)$$

where K and S represent distribution and selection coefficients respectively. The subscript i and j are component i and j in a mixture, and the superscript E and R the extraction and raffinate phases.

One can see from Tables 2–4 that the distribution for benzene generally follows the trend of [EMIM][NTf₂] > Sulfolane >

Table 1
Specifications of chemicals.

Chemical name	CAS No.	Mass fraction purity	Purification method	Final water mass fraction/%	Analysis method
n-Hexane	110-54-3	≥0.995	None	≤0.05	GC ^b and KF ^c
Benzene	71-43-2	≥0.995	None	≤0.01	GC ^b and KF ^c
Ethanol	64-17-5	≥0.997	None	≤0.03	GC ^b and KF ^c
Sulfolane	126-33-0	≥0.990	None	≤0.04	GC ^b and KF ^c
[EMIM][NTf ₂]	174899-82-2	≥0.990 ^a	Vacuum desiccation	≤0.02	KF ^c
[EMIM][EtSO ₄]	342573-75-5	≥0.990 ^a	Vacuum desiccation	≤0.06	KF ^c

Note:

^a The purities reported by supplier.^b GC = Gas chromatography^c KF = Karl Fischer titrator.**Table 2**Experimental LLE Data for the ternary system n-hexane (1) + benzene (2) + sulfolane (3) in mole fraction at different temperatures.^a

T/K	n-Hexane rich phase		Sulfolane rich phase		K_1	K_2	S
	x_1	x_2	x_1	x_2			
303.15	0.8675	0.0668	0.0148	0.0483	0.0171	0.7231	42.38
	0.8190	0.1094	0.0161	0.0780	0.0197	0.7130	36.27
	0.7161	0.2208	0.0256	0.1889	0.0357	0.8555	23.93
	0.6420	0.2808	0.0275	0.2432	0.0428	0.8661	20.22
	0.5566	0.3707	0.0309	0.3000	0.0555	0.8093	14.58
	0.4548	0.4638	0.0533	0.4136	0.1172	0.8918	7.61
	0.3532	0.5326	0.0781	0.4834	0.2211	0.9076	4.10
313.15	0.2511	0.5766	0.1259	0.5395	0.5014	0.9357	1.87
	0.8443	0.0750	0.0147	0.0548	0.0174	0.7307	41.97
	0.8102	0.1119	0.0174	0.0943	0.0215	0.8427	39.24
	0.7050	0.2172	0.0256	0.1755	0.0363	0.8080	22.25
	0.6505	0.2747	0.0312	0.2325	0.0480	0.8464	17.65
	0.5747	0.3610	0.0412	0.3111	0.0717	0.8618	12.02
	0.4507	0.4715	0.0618	0.4231	0.1371	0.8973	6.54
323.15	0.3451	0.5352	0.0789	0.4646	0.2286	0.8681	3.80
	0.2297	0.5697	0.1159	0.5225	0.5046	0.9171	1.82
	0.8008	0.0720	0.0147	0.0556	0.0184	0.7722	42.07
	0.7566	0.1155	0.0240	0.1021	0.0317	0.8840	27.87
	0.6822	0.2014	0.0276	0.1691	0.0405	0.8396	20.75
	0.6253	0.2806	0.0389	0.2518	0.0622	0.8974	14.42
	0.5201	0.3893	0.0543	0.3558	0.1044	0.9139	8.75
	0.4335	0.4715	0.0671	0.4183	0.1548	0.8872	5.73
	0.3174	0.5428	0.0981	0.4663	0.3091	0.8591	2.78
	0.2273	0.5903	0.1537	0.5389	0.6762	0.9129	1.35

^a Standard uncertainty u are $u(T)=0.12\text{K}$ and $u(x)=0.0012$.**Table 3**Experimental LLE Data for the ternary system n-hexane (1) + benzene (2) + [EMIM][EtSO₄] (3) in mole fraction at different temperatures.^a

T/K	n-Hexane rich phase		[EMIM][EtSO ₄] rich phase		K_1	K_2	S
	x_1	x_2	x_1	x_2			
303.15	0.8723	0.1277	0.0081	0.0803	0.0093	0.6288	67.72
	0.8725	0.1275	0.0081	0.0790	0.0093	0.6196	66.74
	0.7261	0.2739	0.0068	0.1199	0.0094	0.4378	46.74
	0.6634	0.3366	0.0058	0.1259	0.0087	0.3740	42.78
	0.5857	0.4143	0.0046	0.1594	0.0079	0.3847	48.99
	0.4264	0.5736	0.0081	0.2259	0.0190	0.3938	20.73
	0.3414	0.6586	0.0061	0.2391	0.0179	0.3630	20.32
313.15	0.2341	0.7659	0.0062	0.3363	0.0265	0.4391	16.58
	0.9225	0.0775	0.0070	0.0382	0.0076	0.4929	64.96
	0.8711	0.1289	0.0040	0.0397	0.0046	0.3080	67.07
	0.7315	0.2685	0.0068	0.1113	0.0093	0.4145	44.59
	0.6225	0.3775	0.0071	0.1640	0.0114	0.4344	38.09
	0.5303	0.4697	0.0091	0.1856	0.0172	0.3951	23.03
	0.4276	0.5724	0.0053	0.2016	0.0124	0.3522	28.41
323.15	0.3455	0.6545	0.0057	0.2463	0.0165	0.3763	22.81
	0.2240	0.7760	0.0056	0.2893	0.0250	0.3728	14.91
	0.9206	0.0794	0.0062	0.0313	0.0067	0.3942	58.53
	0.8709	0.1291	0.0059	0.0505	0.0068	0.3912	57.74
	0.7230	0.2770	0.0058	0.1072	0.0080	0.3870	48.24
	0.6197	0.3803	0.0064	0.1453	0.0103	0.3821	36.99
	0.5190	0.4810	0.0056	0.1727	0.0108	0.3590	33.28
	0.4195	0.5805	0.0052	0.2057	0.0124	0.3543	28.59
	0.3386	0.6614	0.0050	0.2410	0.0148	0.3644	24.67
	0.2204	0.7796	0.0049	0.2764	0.0222	0.3545	15.95

^a Standard uncertainty u are $u(T)=0.12\text{K}$ and $u(x)=0.0012$.

Table 4Experimental LLE Data for the ternary system n-hexane (1) + benzene (2) + [EMIM][NTf₂] (3) in mole fraction at different temperatures.^a

T/K	n-Hexane rich phase		[EMIM][NTf ₂] rich phase		K ₁	K ₂	S
	x ₁	x ₂	x ₁	x ₂			
303.15	0.9845	0.0155	0.0330	0.0211	0.0335	1.3613	40.61
	0.9429	0.0571	0.0421	0.0737	0.0446	1.2907	28.91
	0.7479	0.2521	0.0370	0.2961	0.0495	1.1745	23.74
	0.6564	0.3436	0.0371	0.3704	0.0565	1.0780	19.07
	0.5536	0.4464	0.0356	0.4023	0.0643	0.9012	14.01
	0.4576	0.5424	0.0354	0.5084	0.0774	0.9373	12.12
	0.2927	0.7073	0.0302	0.5854	0.1032	0.8277	8.02
	0.1766	0.8234	0.0235	0.6484	0.1331	0.7875	5.92
	0.9831	0.0169	0.0428	0.0195	0.0435	1.1538	26.50
	0.9546	0.0454	0.0458	0.0579	0.0480	1.2753	26.58
313.15	0.7396	0.2604	0.0403	0.2945	0.0545	1.1310	20.76
	0.6331	0.3669	0.0344	0.3475	0.0543	0.9471	17.43
	0.5485	0.4515	0.0392	0.4400	0.0715	0.9745	13.64
	0.4477	0.5523	0.0327	0.4980	0.0730	0.9017	12.34
	0.2771	0.7229	0.0310	0.5776	0.1119	0.7990	7.14
	0.1681	0.8319	0.0224	0.6415	0.1333	0.7711	5.79
	0.9845	0.0155	0.0485	0.0204	0.0493	1.3161	26.72
323.15	0.9503	0.0497	0.0457	0.0610	0.0481	1.2274	25.52
	0.7421	0.2579	0.0427	0.2818	0.0575	1.0927	18.99
	0.6105	0.3895	0.0397	0.3565	0.0650	0.9153	14.07
	0.5481	0.4519	0.0373	0.4229	0.0681	0.9358	13.75
	0.4479	0.5521	0.0385	0.4862	0.0860	0.8806	10.24
	0.2847	0.7153	0.0320	0.5794	0.1124	0.8100	7.21
	0.1559	0.8441	0.0247	0.6395	0.1584	0.7576	4.78

^a Standard uncertainty u are u(T)=0.12K and u(x)=0.0012.**Table 5**Experimental LLE Data for the Pseudoternary System n-hexane (1) + benzene (2) + Mixed ILs ([EMIM][EtSO₄]: [EMIM][NTf₂] = 1: 9) (3) in mole fraction at different temperatures.^a

T/K	n-Hexane rich phase		[EMIM][NTf ₂] rich phase		K ₁	K ₂	S
	x ₁	x ₂	x ₁	x ₂			
303.15	0.9229	0.0771	0.0252	0.0756	0.0273	0.9805	35.91
	0.8784	0.1216	0.0221	0.1054	0.0252	0.8668	34.45
	0.7646	0.2354	0.0223	0.1908	0.0292	0.8105	27.79
	0.6657	0.3343	0.0249	0.2652	0.0374	0.7933	21.21
	0.5758	0.4242	0.0253	0.3098	0.0439	0.7303	16.62
	0.5460	0.4540	0.0304	0.4186	0.0557	0.9220	16.56
	0.3827	0.6173	0.0205	0.4984	0.0536	0.8074	15.07
	0.2924	0.7076	0.0200	0.5597	0.0684	0.7910	11.56
	0.9276	0.0724	0.0222	0.0603	0.0239	0.8329	34.80
	0.8859	0.1141	0.0248	0.1077	0.0280	0.9439	33.72
313.15	0.7998	0.2002	0.0254	0.2014	0.0318	1.0060	31.68
	0.7198	0.2802	0.0217	0.2263	0.0301	0.8076	26.79
	0.6162	0.3838	0.0269	0.3724	0.0437	0.9703	22.23
	0.5425	0.4575	0.0249	0.3899	0.0459	0.8522	18.57
	0.4386	0.5614	0.0309	0.4751	0.0705	0.8463	12.01
	0.3127	0.6873	0.0290	0.5667	0.0927	0.8245	8.89
	0.9345	0.0655	0.0294	0.0633	0.0315	0.9664	30.72
	0.8936	0.1064	0.0324	0.1036	0.0363	0.9737	26.85
	0.8106	0.1894	0.0336	0.2102	0.0415	1.1098	26.77
	0.7074	0.2926	0.0350	0.2613	0.0495	0.8930	18.05
323.15	0.6255	0.3745	0.0372	0.3689	0.0595	0.9850	16.56
	0.5377	0.4623	0.0341	0.3957	0.0634	0.8559	13.50
	0.4547	0.5453	0.0335	0.4729	0.0737	0.8672	11.77
	0.3082	0.6918	0.0296	0.5643	0.0960	0.8157	8.49

^a Standard uncertainty u are u(T)=0.12K and u(x)=0.0012.

[EMIM][EtSO₄], while vice versa for selections. However, an attention should be paid that the distributions in [EMIM][NTf₂] systems were not always greater than ones with sulfolane extractant, and the selections in the former not absolutely lower than ones in the latter, which depends on the concentration of benzene in the original solution. The distributions of benzene in sulfolane mixture would be slightly greater than ones of [EMIM][NTf₂] while the corresponding selections of sulfolane would tend to 1, when the benzene composition in original mixture is enough larger. To improve the separation performance of pure ILs, ones of IL mixtures were tested through experimental measurement, as shown

in Tables 5 and 6, where the mixing ratios of [EMIM][EtSO₄] to [EMIM][NTf₂] were 1:9 and 9:1 in mole respectively. It can be seen that the separation performance can well be balanced through mixing the two ILs. The selections for benzene can be effectively improved when a small certain of [EMIM][EtSO₄] were mixed into [EMIM][NTf₂] although the slightly lower distributions were observed. But it is still greater than ones with sulfolane extractant. The experimental ternary liquid-liquid phase diagrams for all systems at different temperature have been depicted in Figures S1-S5 in supporting information (SI), in which the literature data at 313.2 K for the related systems [11,16,37] were also contained. The

Table 6

Experimental LLE Data for the Pseudoternary System n-hexane (1) + benzene (2) + Mixed ILs ([EMIM][EtSO₄]: [EMIM][NTf₂] = 9: 1) (3) in mole fraction at different temperatures.^a

T/K	n-Hexane rich phase		[EMIM]NTf ₂ rich phase		K ₁	K ₂	S
	x ₁	x ₂	x ₁	x ₂			
303.15	0.9129	0.0871	0.0104	0.0515	0.0114	0.5919	52.03
	0.8715	0.1285	0.0102	0.0698	0.0117	0.5427	46.38
	0.7423	0.2577	0.0119	0.1333	0.0160	0.5172	32.37
	0.6550	0.3450	0.0119	0.1825	0.0182	0.5289	29.08
	0.4658	0.5342	0.0107	0.2700	0.0230	0.5053	21.96
	0.3620	0.6380	0.0114	0.3025	0.0315	0.4742	15.03
	0.2405	0.7595	0.0104	0.3462	0.0431	0.4559	10.58
	0.9249	0.0751	0.0100	0.0423	0.0108	0.5631	52.02
313.15	0.8787	0.1213	0.0115	0.0697	0.0131	0.5744	43.90
	0.7396	0.2604	0.0123	0.1395	0.0166	0.5358	32.25
	0.6528	0.3472	0.0130	0.1797	0.0199	0.5174	26.00
	0.5611	0.4389	0.0126	0.2177	0.0224	0.4961	22.14
	0.4600	0.5400	0.0125	0.2590	0.0272	0.4797	17.62
	0.3644	0.6356	0.0107	0.2912	0.0294	0.4582	15.59
	0.2462	0.7538	0.0110	0.3413	0.0448	0.4527	10.10
	0.9227	0.0773	0.0110	0.0456	0.0119	0.5901	49.55
323.15	0.8799	0.1201	0.0111	0.0663	0.0126	0.5519	43.87
	0.7293	0.2708	0.0120	0.1390	0.0165	0.5134	31.16
	0.6362	0.3638	0.0127	0.1878	0.0200	0.5162	25.84
	0.5449	0.4551	0.0115	0.2274	0.0211	0.4997	23.67
	0.4527	0.5473	0.0112	0.2684	0.0247	0.4905	19.84
	0.3556	0.6444	0.0110	0.3032	0.0310	0.4706	15.17
	0.2492	0.7508	0.0112	0.3467	0.0451	0.4618	10.25

^a Standard uncertainty u are u(T)=0.12K and u(x)=0.0012.

consistent results with pure IL extractants between this work and literatures [11,16] were obtained although the analysis methods in both were different. For sulfolane system, both distributions and selections of benzene from literature [37] were lower than ones measured in this work, which might be induced by experimental measurement or analysis methods and so on.

3.2. Correlations of experimental LLE data

The non-random two liquid (NRTL) model was employed to calculate the LLE properties since the NRTL is applicable to partial and all mutually soluble systems. Many researches [11,12,14–16,18,19,34,38–40] have used this model to reproduce experimental LLE data with ILs and all of them achieved good results. The NRTL model is expressed as follows:

$$\ln \gamma_i = \frac{\sum_j \tau_{ji} G_{ji} x_j}{\sum_k G_{ki} x_k} + \sum_j \frac{G_{ij} x_j}{\sum_k G_{kj} x_k} \left(\tau_{ij} - \frac{\sum_k \tau_{kj} G_{kj} x_k}{\sum_k G_{kj} x_k} \right) \quad (1a)$$

$$G_{ji} = \exp(-\alpha_{ji} \tau_{ji}) \quad (2a)$$

$$\tau_{ji} = a_{ji} + \frac{b_{ji}}{T} \quad (3)$$

where γ is the activity coefficient, x the mole fraction in the liquid phase, T the system temperature, α_{ji} non-random parameters, a_{ji} and b_{ji} binary interaction parameter fitted from experimental data. The subscripts i and j delegate component i and j .

The binary interaction parameters were estimated by using a procedure which is based on the modified Simplex method with the following objective function:

$$OF = \sqrt{\frac{\sum_k^N \sum_i^C (x_{ik}^E \gamma_{ik}^E - x_{ik}^R \gamma_{ik}^R)^2}{NC}} \quad (4)$$

where x_{ik} and x_{ik} are the experimental equilibrium compositions of component i ; γ_{ik} and γ_{ik} the activity coefficient of component i in two equilibrium phases; C is the number of components; N the number of experiment points. The average absolute deviation was

calculated by:

$$AAD = \frac{\sum_k^N \sum_i^{2(C-1)} |x_{ik}^{cal} - x_{ik}^{exp}|}{2N(C-1)} \quad (7)$$

According to the above calculation methodology, the binary interaction parameters in NRTL model for the investigated systems were obtained based on the experimental data and listed in Table 7. Firstly, the binary interaction parameters between n-hexane and benzene were obtained based on vapor-liquid equilibrium data from literature [41]. After that, the cross parameters of between n-hexane or benzene and different extractants were calculated according to experimental data measured in this work. At last, the NRTL model parameters between [EMIM][NTf₂] and [EMIM][EtSO₄] were determined with the experimental data of the two quaternary mixtures and all other fixed binary interaction parameters. It can be seen that the NRTL model can satisfactorily reproduce experimental data. The theoretical LLE results were listed in Tables S1–S5 in SI, in which the theoretical equilibrium compositions, temperatures and distribution and selection coefficients were included. The theoretical results were also illustrated in Figures S1–S5 and good agreements between experimental (in the Supplementary Materials) and theoretical results were generally obtained, although there were several points showing somewhat big deviation. Furthermore, the fitting binary parameters for all ternary systems in this work were checked according to the Gibbs theorem with the method provided in literature [42]. It was found that the obtained binary parameters in this work were reasonable.

In Figure 1, the comparisons of molar distribution and selection coefficients among five cases were depicted, where the x_a represents the average composition of benzene in raffinate and extraction phases. It can be seen that the [EMIM][NTf₂] and [EMIM][EtSO₄] respectively capture higher distribution and selection coefficients compared to sulfolane extractant. With the change of benzene composition in mixtures, a crossing phenomenon between [EMIM][NTf₂] and sulfolane cases were observed. Generally, the [EMIM][NTf₂] could be considered as a potential alternative extractant for industrial application in aromatics extraction. Further-

Table 7
The binary interaction in NRTL model for investigated mixtures.

Component <i>i</i>	Component <i>j</i>	a_{ij}	a_{ji}	b_{ij}	b_{ji}	α_{ij}	AAD
n-Hexane (1) + Benzene (2)							
n-Hexane	Benzene	0.0000	0.0000	-72.7737	278.4858	0.3	See Note
n-Hexane (1) + Benzene (2) + Sulfolane (3)							
n-Hexane	Sulfolane	-5.3273	3.9137	2262.5267	-41.9709	0.3	0.0124
Benzene	Sulfolane	12.3898	-2.8840	-2977.6393	686.9883	0.3	
n-Hexane (1) + Benzene (2) + [EMIM][EtSO ₄] (3)							
n-Hexane	[EMIM][EtSO ₄]	9.0224	5.1002	198.4018	-596.7398	0.16	0.0064
Benzene	[EMIM][EtSO ₄]	9.3758	3.1088	577.6041	-868.6228	0.22	
n-Hexane (1) + Benzene (2) + [EMIM][NTf ₂] (3)							
n-Hexane	[EMIM][NTf ₂]	11.9011	-1.4103	-509.0579	935.9871	0.17	0.0052
Benzene	[EMIM][NTf ₂]	8.9730	0.5187	473.1898	-198.8887	0.33	
n-Hexane (1) + Benzene (2) + [EMIM][NTf ₂] (3) + [EMIM][EtSO ₄] (4)							
[EMIM][NTf ₂]	[EMIM][EtSO ₄]	-0.4372	-0.1975	6.5110	142.5060	0.99	0.0124

Note: The average absolute deviations of pressure and composition in vapor phase are 0.21 kPa and 0.004.

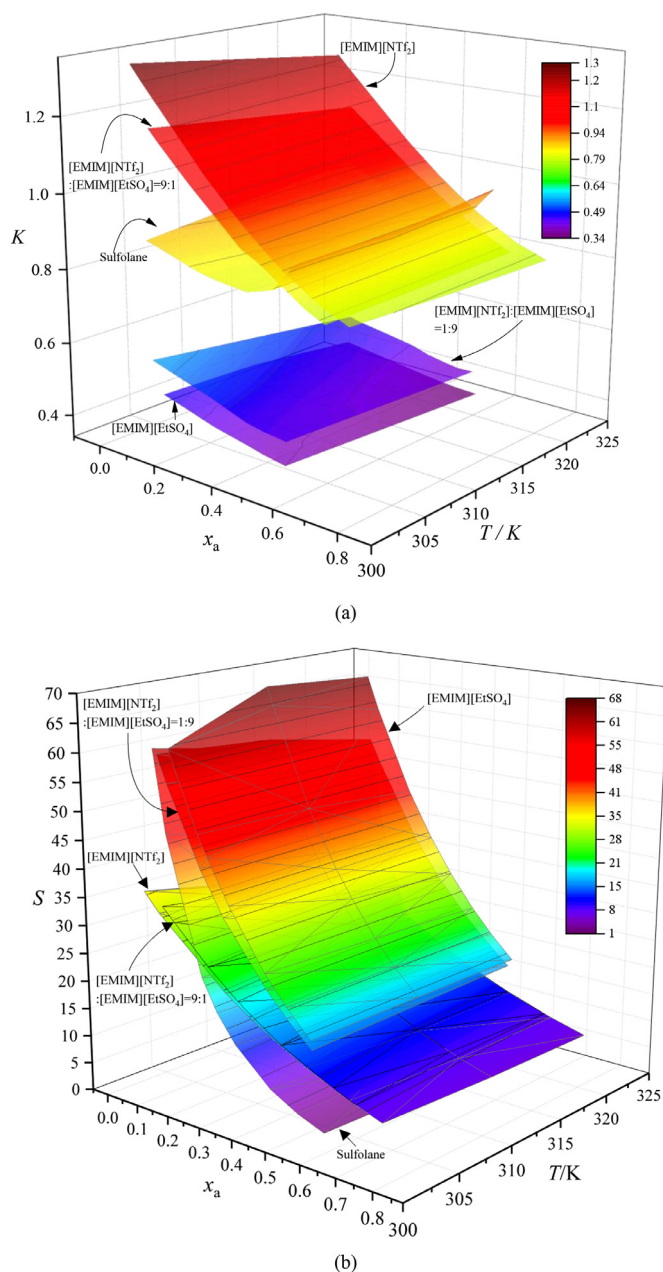


Figure 1. Comparisons of distribution (a) and selection (b) coefficients among different cases. (x_a is the mole fraction of benzene, K partition coefficient and S selectivity).

more, the separation performance of [EMIM][NTf₂] can be effectively improved by adding small amount of [EMIM][EtSO₄].

4. Conclusions

The LLE data for three ternary and two quaternary mixtures containing sulfolane, [EMIM][NTf₂] and [EMIM][EtSO₄] extractants were experimentally determined at 303.15, 313.15 and 323.15 K and atmospheric pressure with self-made glass equilibrium cell. As for pure extractant in the separation of n-hexane and benzene, the ionic liquids of [EMIM][NTf₂] and [EMIM][EtSO₄] respectively capture higher distribution and selection coefficients generally, compared to sulfolane solvent. The overall separation performance of [EMIM][NTf₂] would be comparable to one of sulfolane. Meanwhile, the separation ability of [EMIM][NTf₂] can be significantly improved by adding a small amount of [EMIM][EtSO₄]. To provide the theoretical parameters for the process design, simulation and optimization, the NRTL model was employed and successful calculations were made. In general, the [EMIM][NTf₂] or the ionic liquid mixture of [EMIM][NTf₂] and [EMIM][EtSO₄] would be a good potential representative extractant in industry for aromatics extraction. Appendix A. Supplementary data related to this article can be found at <https://doi.org/10.1016/j.fluid.2020.112882>.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] G.W. Meindersma, A.B. de Haan, Conceptual process design for aromatic/aliphatic separation with ionic liquids, *Chemical Engineering Research and Design* 86 (2008) 745–752.
- [2] T. Zhou, Z. Wang, Y. Ye, L. Chen, J. Xu, Z. Qi, Deep Separation of Benzene from Cyclohexane by Liquid Extraction Using Ionic Liquids as the Solvent, *Industrial & Engineering Chemistry Research* 51 (2012) 5559–5564.
- [3] Z. Ren, M. Wang, Y. Li, Z. Zhou, F. Zhang, W. Liu, Selective Separation of Benzene/n-Hexane with Ester-Functionalized Ionic Liquids, *Energy & Fuels* 31 (2017) 6598–6606.
- [4] C.-F. Kao, W.-C. Lin, Liquid–Liquid Equilibria of the Ternary Systems Dodecane + Butylbenzene + Sulfolane, Dodecane + 1,4-Diisopropylbenzene + Sulfolane, and Dodecane + Octylbenzene + Sulfolane, *Journal of Chemical & Engineering Data* 44 (1999) 338–342.

- [5] T.M. Letcher, G.G. Redhi, S.E. Radloff, U. Domańska, Liquid–Liquid Equilibria of the Ternary Mixtures with Sulfolane at 303.15 K, *Journal of Chemical & Engineering Data* 41 (1996) 634–638.
- [6] W.-C. Lin, N.-H. Kao, Liquid–Liquid Equilibria of Octane + (Benzene or Toluene or m-Xylene) + Sulfolane at 323.15, 348.15, and 373.15 K, *Journal of Chemical & Engineering Data* 47 (2002) 1007–1011.
- [7] I. Ashour, S.I. Abu-Eishah, Liquid–Liquid Equilibria of Ternary and Six-Component Systems Including Cyclohexane, Benzene, Toluene, Ethylbenzene, Cumene, and Sulfolane at 303.15 K, *Journal of Chemical & Engineering Data* 51 (2006) 1717–1722.
- [8] A.B. Pereiro, J.M.M. Araujo, J.M.S.S. Esperanca, I.M. Marrucho, L.P.N. Rebelo, Ionic liquids in separations of azeotropic systems - A review, *J. Chem. Thermodyn.* 46 (2012) 2–28.
- [9] A. Arce, M.J. Earle, S.P. Katdare, H. Rodríguez, K.R. Seddon, Application of mutually immiscible ionic liquids to the separation of aromatic and aliphatic hydrocarbons by liquid extraction: a preliminary approach, *Phys Chem Chem Phys* 10 (2008) 2538–2542.
- [10] A. Arce, M.J. Earle, H. Rodríguez, K.R. Seddon, Separation of benzene and hexane by solvent extraction with 1-alkyl-3-methylimidazolium bis((trifluoromethyl)sulfonyl)amide ionic liquids: effect of the alkyl-substituent length, *J Phys Chem B* 111 (2007) 4732–4736.
- [11] A. Arce, M.J. Earle, H. Rodríguez, K.R. Seddon, Separation of aromatic hydrocarbons from alkanes using the ionic liquid 1-ethyl-3-methylimidazolium bis((trifluoromethyl) sulfonyl)amide, *Green Chem* 9 (2007) 70–74.
- [12] A. Arce, M.J. Earle, H. Rodríguez, K.R. Seddon, A. Soto, Bis((trifluoromethyl)sulfonyl)amide ionic liquids as solvents for the extraction of aromatic hydrocarbons from their mixtures with alkanes: effect of the nature of the cation, *Green Chemistry* 11 (2009) 365–372.
- [13] U. Domańska, A. Pobudkowska, F. Eckert, Liquid–liquid equilibria in the binary systems (1,3-dimethylimidazolium, or 1-butyl-3-methylimidazolium methylsulfate + hydrocarbons), 8 (2006) 268–276.
- [14] I. Domínguez, N. Calvar, E. Gómez, Á. Domínguez, Liquid–Liquid Extraction of Aromatic Compounds from Cycloalkanes Using 1-Butyl-3-methylimidazolium Methylsulfate Ionic Liquid, *Journal of Chemical & Engineering Data* 58 (2013) 189–196.
- [15] I. Domínguez, E.J. González, R. González, A.n. Domínguez, Extraction of Benzene from Aliphatic Compounds Using Commercial Ionic Liquids as Solvents: Study of the Liquid–Liquid Equilibrium at $T = 298.15$ K, *Journal of Chemical & Engineering Data* 56 (2011) 3376–3383.
- [16] J. García, A. Fernández, J.S. Torrecilla, M. Oliet, F. Rodríguez, Liquid–liquid equilibria for {hexane+benzene+1-ethyl-3-methylimidazolium ethylsulfate} at (298.2, 313.2 and 328.2)K, *Fluid Phase Equilibria* 282 (2009) 117–120.
- [17] S. García, M. Larriba, J. García, J.S. Torrecilla, F. Rodríguez, Liquid–Liquid Extraction of Toluene from Heptane Using 1-Alkyl-3-methylimidazolium Bis(trifluoromethylsulfonyl)imide Ionic Liquids, *Journal of Chemical & Engineering Data* 56 (2011) 113–118.
- [18] S. García, M. Larriba, J.n. García, J.S. Torrecilla, F. Rodríguez, 1-Alkyl-2,3-dimethylimidazolium Bis(trifluoromethylsulfonyl)imide Ionic Liquids for the Liquid–Liquid Extraction of Toluene from Heptane, *Journal of Chemical & Engineering Data* 56 (2011) 3468–3474.
- [19] E.J. González, N. Calvar, I. Domínguez, Á. Domínguez, (Liquid+liquid) equilibrium data for the ternary systems (cycloalkane+ethylbenzene+1-ethyl-3-methylimidazolium ethylsulfate) at $T = 298.15$ K and atmospheric pressure, *The Journal of Chemical Thermodynamics* 43 (2011) 725–730.
- [20] E.J. González, N. Calvar, E. Gómez, Á. Domínguez, Separation of Benzene from Linear Alkanes (C6–C9) Using 1-Ethyl-3-Methylimidazolium Ethylsulfate at $T = 298.15$ K, *Journal of Chemical & Engineering Data* 55 (2010) 3422–3427.
- [21] E.J. González, N. Calvar, B. González, Á. Domínguez, Measurement and correlation of liquid–liquid equilibria for ternary systems {cyclooctane+aromatic hydrocarbon+1-ethyl-3-methylpyridinium ethylsulfate} at $T = 298.15$ K and atmospheric pressure, *Fluid Phase Equilibria* 291 (2010) 59–65.
- [22] E.J. González, N. Calvar, B. González, Á. Domínguez, Liquid–Liquid Equilibrium for Ternary Mixtures of Hexane + Aromatic Compounds + [EMPy][ESO4] at $T = 298.15$ K, *Journal of Chemical & Engineering Data* 55 (2010) 633–638.
- [23] E.J. González, I. Domínguez, B. González, J. Canosa, Liquid–liquid equilibria for ternary systems of {cyclohexane+aromatic compounds+1-ethyl-3-methylpyridinium ethylsulfate}, *Fluid Phase Equilibria* 296 (2010) 213–218.
- [24] M. Larriba, J. de Riva, P. Navarro, D. Moreno, N. Delgado-Mellado, J. García, V.R. Ferro, F. Rodríguez, J. Palomar, COSMO-based/Aspen Plus process simulation of the aromatic extraction from pyrolysis gasoline using the ([4empy][NTf2] + [emim][DCA]) ionic liquid mixture, *Separation and Purification Technology* 190 (2018) 211–227.
- [25] M. Larriba, P. Navarro, N. Delgado-Mellado, V. Stanisci, J. García, F. Rodríguez, Separation of aromatics from n-alkanes using tricyanomethanide-based ionic liquids: Liquid–liquid extraction, vapor–liquid separation, and thermophysical characterization, *Journal of Molecular Liquids* 223 (2016) 880–889.
- [26] M. Larriba, P. Navarro, J. García, F. Rodríguez, Selective extraction of toluene from n-heptane using emim SCN and bmim SCN ionic liquids as solvents, *Journal of Chemical Thermodynamics* 79 (2014) 266–271.
- [27] M. Larriba, P. Navarro, J. García, F. Rodríguez, Liquid–Liquid Extraction of Toluene from Heptane Using [emim][DCA], [bmim][DCA], and [emim][TCM] Ionic Liquids, *Industrial & Engineering Chemistry Research* 52 (2013) 2714–2720.
- [28] M. Larriba, P. Navarro, J. García, F. Rodríguez, Liquid–Liquid Extraction of BTEX from Reformer Gasoline Using Binary Mixtures of [4empy][Tf2N] and [emim][DCA] Ionic Liquids, *Energy & Fuels* 28 (2014) 6666–6676.
- [29] G.W. Meindersma, A.B. De Haan, Cyano-containing ionic liquids for the extraction of aromatic hydrocarbons from an aromatic/aliphatic mixture, *Science China Chemistry* 55 (2012) 1488–1499.
- [30] G.W. Meindersma, A.R. Hansmeier, A.B. de Haan, Ionic Liquids for Aromatics Extraction. Present Status and Future Outlook, *Industrial & Engineering Chemistry Research* 49 (2010) 7530–7540.
- [31] G.W. Meindersma, A. Podt, A.B. de Haan, Ternary Liquid–Liquid Equilibria for Mixtures of an Aromatic + an Aliphatic Hydrocarbon + 4-Methyl-N-butylpyridinium Tetrafluoroborate, *Journal of Chemical & Engineering Data* 51 (2006) 1814–1819.
- [32] G.W. Meindersma, A.J.G. Podt, A.B. de Haan, Ternary liquid–liquid equilibria for mixtures of toluene+n-heptane+an ionic liquid, *Fluid Phase Equilibria* 247 (2006) 158–168.
- [33] R. Maduro, M. Aznar, Liquid–liquid equilibrium of ternary systems 1-octyl-3-methylimidazolium hexafluorophosphate + aromatic + aliphatic hydrocarbons, *Fluid Phase Equilibria* 296 (2010) 88–94.
- [34] M.S. Al-Tuwaim, K.H.A.E. Alkhalidi, M.S. Fandary, A.S. Al-Jimaz, Extraction of propylbenzene or butylbenzene from dodecane using 4-methyl-N-butylpyridinium tetrafluoroborate, [mebupy][BF4], as an ionic liquid at different temperatures, *Journal of Chemical Thermodynamics* 43 (2011) 1804–1809.
- [35] S. García, M. Larriba, A. Casas, J. García, F. Rodríguez, Separation of Toluene and Heptane by Liquid–Liquid Extraction Using Binary Mixtures of the Ionic Liquids 1-Butyl-4-methylpyridinium Bis(trifluoromethylsulfonyl)imide and 1-Ethyl-3-methylimidazolium Ethylsulfate, *Journal of Chemical & Engineering Data* 57 (2012) 2472–2478.
- [36] M. Larriba, P. Navarro, J. García, F. Rodríguez, Liquid–Liquid Extraction of Toluene from n-Alkanes using ([4empy][Tf2N] + [emim][DCA]) Ionic Liquid Mixtures, *Journal of Chemical & Engineering Data* 59 (2014) 1692–1699.
- [37] J. Mahmoudi, M.N. Lotfollahi, (Liquid+liquid) equilibria of (sulfolane+benzene+n-hexane), (N-formylmorpholine+benzene+n-hexane), and (sulfolane+N-formylmorpholine+benzene+n-hexane) at temperatures ranging from (298.15 to 318.15)K: Experimental results and correlation, *The Journal of Chemical Thermodynamics* 42 (2010) 466–471.
- [38] N. Calvar, I. Domínguez, E. Gomez, J. Palomar, A. Domínguez, Evaluation of ionic liquids as solvent for aromatic extraction: Experimental, correlation and COSMO-RS predictions, *Journal of Chemical Thermodynamics* 67 (2013) 5–12.
- [39] S. Corderi, N. Calvar, E. Gomez, A. Domínguez, Capacity of ionic liquids [EMim][NTf2] and [EMPy][NTf2] for extraction of toluene from mixtures with alkanes: Comparative study of the effect of the cation, *Fluid Phase Equilibria* 315 (2012) 46–52.
- [40] M. Ebrahimi, A.N. Ahmadi, A.A. Safekordi, F. Fateminasab, A. Mehdizadeh, Liquid–Liquid Equilibrium Data for {Heptane + Aromatic + 1-(2-Hydroxyethyl)-3-methylimidazolium Bis(trifluoromethylsulfonyl)imide ([hemim][NTf2])} Ternary Systems, *Journal of Chemical & Engineering Data* 59 (2014) 197–204.
- [41] I.P.C. Li, Y.W. Wong, S.D. Chang, B.C.Y. Lu, vapor–liquid equilibria in systems n-hexane-benzene and n-pentane-toluene, *J. Chem Eng. Data* 17 (1972) 492–498.
- [42] A. Marcilla, J.A. Reyes-Labarta, M.M. Olaya, Should we trust all the published LLE correlation parameters in phase equilibria? Necessity of their assessment prior to publication, *Fluid Phase Equilibria* 433 (2017) 243–252.